

1.4 Applying the Laws of Exponents

Lesson Introduction

 Benchmark	MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.			 Learning Target	- I can apply the Laws of Exponents to write equivalent numerical expressions. - I can evaluate square roots and cube roots.	
 Benchmark Coherence	Building On			Working Toward		
	MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.			N/A		
 Essential Questions	- How can the Laws of Exponents help me to evaluate an expression? - What are the key differences between a cube root and a square root?					
 Lesson Narrative	In this lesson, students will revisit the Laws of Exponents, square roots, and cube roots. Students will review the laws of exponents and then transition to working with square and cube roots. They will have the opportunity to practice evaluating expressions using the Laws of Exponents and knowledge of square and cube roots.					
 Component Summary	1.4.1 Warm-Up (~5 min.)	Toothpick Puzzle Challenge (<i>SE p. 23</i>)	1.4.3 Guided Instruction (15 min.)	Extending knowledge of square roots and cube roots (<i>SE p. 24</i>)		
	1.4.2 Refresher (10 min.)	Refresher on the Laws of Exponents (<i>SE p. 23</i>)	1.4.4 Your Turn! (10 min.)	Collaboratively determine the values of square roots and rewriting expressions using the Laws of Exponents (<i>SE p. 25</i>)		
	1.4.5 Wrap-Up (~5 min.)	Cool down that compares two expressions using inequality symbols. (<i>SE p. 26</i>)				
 Resources	Glossary		Materials	Additional Preparation		
	<u>Key Terms:</u> - Principal square root <u>Additional Terms:</u> <table><tr><td> - Cube Root - Square Root - Product of Powers - Quotient of Powers - Power of a Power </td><td> - Power of a Product - Power of a Quotient - Negative Exponents - Identity Exponents - Zero Exponent </td></tr></table>		- Cube Root - Square Root - Product of Powers - Quotient of Powers - Power of a Power	- Power of a Product - Power of a Quotient - Negative Exponents - Identity Exponents - Zero Exponent	N/A	(Optional) consider making an anchor chart or having materials to make a tabletop chart for students to reference the Laws of Exponents.
- Cube Root - Square Root - Product of Powers - Quotient of Powers - Power of a Power	- Power of a Product - Power of a Quotient - Negative Exponents - Identity Exponents - Zero Exponent					

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mistake the definition of a square as double a number. - Students may mistake the definition of a cube as triple a number. - Students may mistake a cube root for a square root.	Students may have varying entry points when accessing this content. Depending on the math courses they have taken prior to Algebra 1, students may have limited working knowledge of exponent laws.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	True or False When presenting the Laws of Exponent table in 1.4.2 Refresher, consider implementing a True or False protocol. Present each answer and ask students if the resulting expression is true or false and allow them to justify why they think that is true or false.	MA.K12.MTR.5.1: Patterns and Structure Prior grade level standards introduced students to the Laws of Exponents. Students will extend that knowledge to include rational exponents. Students can use 1.4.2 Refresher to establish the patterns and structures of the Laws of Exponents.
	Student performance during 1.4.2 Refresher can provide valuable differentiation that may be necessary based on the student's point of entry to the Laws of Exponents. If students struggle to understand, consider differentiating for those students in a later activity.	
	1.4.4 Your Turn! provides a table for students to complete about the Laws of Exponents. Consider creating a tabletop chart or anchor chart that provides references for students to use as they work on this activity.	
	The 1.4.5 Wrap-Up for this activity is a cool down that will provide a formative assessment for students to determine their level of understanding of the content of this lesson. Student performance on this activity can provide data for potential differentiation needs.	
Lesson Notes:		

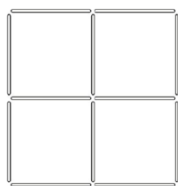
1.4.1 Warm-Up: Toothpick Squares

Component Overview: Students will remove two toothpicks to create two squares with two different sizes. This activity can be done independently.



1.4.1 Warm-Up: Toothpick Squares

1. Make two squares of different sizes by removing two toothpicks.



Answers will vary.



Consider providing actual toothpicks. Being able to manipulate the toothpicks themselves, instead of doing it with just the visual from the book, can be a beneficial way for kinesthetic learners to connect to the activity and better complete the task in a way that makes sense to them.



Consider allowing students to share out their methods for solving this problem.

1.4.2 Refresher: Laws of Exponents

Component Overview: Students will receive a refresher instructional lesson on the Laws of Exponents. For this activity, review the exponent law with a variable, explaining what is happening with this law. Then, show students how it applies to integer values and ask students what the equivalent expressions column would look like, providing them the correct answer for their notes. Repeat this process for each exponent law.



1.4.2 Refresher: Laws of Exponents

- Using the given exponent law, complete the table by writing an equivalent exponential expression.

Exponent Law	Expression	Equivalent Expression
Product of Powers $a^m \cdot a^n = a^{m+n}$	$6^5 \cdot 6^2$	6^7
Quotient of Powers $\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^{14}}{5^2}$	5^{12}
Power of a Power $(a^m)^n = a^{m \cdot n}$	$(4^{10})^3$	4^{30}
Power of a Product $(ab)^m = a^m \cdot b^m$	$(5 \cdot 2)^4$	$5^4 \cdot 2^4$
Power of a Quotient $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\left(\frac{3}{11}\right)^2$	$\frac{3^2}{11^2}$
Negative Exponents $a^{-1} = \frac{1}{a}$	9^{-1}	$\frac{1}{9}$
$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$	$\left(\frac{5}{7}\right)^{-1}$	$\frac{7}{5}$
$\left(\frac{a}{b}\right)^{-m} = \frac{b^m}{a^m}$	$\left(\frac{13}{4}\right)^{-3}$	$\frac{4^3}{13^3}$
Identity Exponent $a^1 = a$	572^1	572
Zero Exponent $a^0 = 1$	572^0	1



These laws will be necessary to be successful in the rest of this unit. Consider creating an anchor chart of this information and posting it in a location in the classroom where students can easily reference this as they need throughout the lesson and the unit.



Students may have varying entry points when accessing this content. Depending on the math courses they have taken prior to Algebra, students may have limited working knowledge of these exponent laws. Therefore, this activity will be a refresher for some and new content for others.



How can knowledge of these exponent laws help simplify the evaluation process?

1.4.3 Guided Instruction: Building on Square Roots and Cube Roots

Component Overview: Students will receive guided instruction on expanding their knowledge of square roots and cube roots. For this activity, walk students through the instruction provided in this section. Consider stopping whenever an answer is required to elicit student response, but ensuring they have the correct answer for their notes.



1.4.3 Guided Instruction: Building on Square Roots and Cube Roots

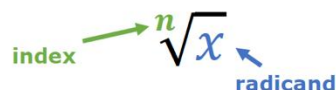
To square a number means to

- multiply it by itself.
- multiply it by 2.

$$7^2 = 7 \cdot 7 = 49$$

$$(-7)^2 = (-7) \cdot (-7) = 49$$

The inverse of squaring a number is taking the square root ($\sqrt{\quad}$).



To evaluate $\sqrt[3]{25}$, also written as $\sqrt[3]{25}$, ask, "What times itself equals 25?"

$$5 \cdot 5 = 25 \text{ or } -5 \cdot -5 = 25$$

Therefore, $\sqrt{25}$ is 5 and -5 .



All radicands greater than or equal to 0, $x \geq 0$, have two square roots, a positive and a negative. The positive square root is referred to as the **principal square root**.



- How can the value of a number squared differ if the understanding of what the square means is "multiply it by 2" instead of "multiply it by itself"?
- How might 2^2 be confusing when discussing the definition?



Consider using this example as a potential anchor chart or visual reference for students to recall. While they may not fluently use these terms in this lesson, they will need to use them in future lessons. Use these terms when doing a think-aloud or providing instruction to reinforce the terms.

For example, say "index of 3" when describing "cube roots" so students can continue to make the connections of a cube root and its index value of 3.



Consider using a language activity for this section. For example, write $\sqrt{25}$ on the board. For each student, ask for one piece of information about the radical. This can be the index, a square root, the principal square root, or what it means. Ask one student for one, another for another, and repeat for all students. You will repeat what you're asking for, but it will help students connect the vocabulary to the content.

1.4.3 Guided Instruction: Building on Square Roots and Cube Roots (continued)

The square root of 16, written as $\sqrt{16}$, is 4 and -4 because both 4^2 and $(-4)^2$ are equal to 16.

To cube a number means to

- multiply the number by 3.
- triple the number.
- multiply the number by itself three times.

$$6^3 = 6 \cdot 6 \cdot 6 = 216 \quad (-6)^3 = (-6) \cdot (-6) \cdot (-6) = -216$$

The cube root of a number is a value that when multiplied by itself three times gives that number.

$$27 = 3^3 \text{ so } \sqrt[3]{27} = 3$$

$$-27 = (-3)^3 \text{ so } \sqrt[3]{-27} = -3$$

The cube root of 64, written as $\sqrt[3]{64}$, is 4 because 4^3 is 64.

$$4 \cdot 4 \cdot 4 = 64$$



- How can the value of a number cubed differ if the understanding of what the cube means is “multiply it by 3” or “triple the number” instead of “multiply it by itself”?
- How might 3^3 be confusing when discussing the definition?



- Why do you think the result of a negative number squared is positive, but when it’s cubed it would be negative?
- What if a number was raised to the 4^{th} or 9^{th} power?



For a challenge, have students determine a number that would be the same when squared as it is when cube rooted. Have them share how they arrived at that answer and if changing the sign of that number would cause it to no longer be true.

1.4.4 Your Turn!

Component Overview: Students will work on writing expressions using the Laws of Exponents. For this activity, begin by releasing students to do question 1 on their own, and then bring the class back together to review the answers. Consider having students work with their partner to complete the table in question 2, then bring the class back together to review the answers. Then, students will work independently on questions 3 and 4.



1.4.4 Your Turn!

1. Find the value(s) of each.

a. $\sqrt{121} = -11 \text{ and } 11$

b. $\sqrt[3]{125} = 5$

2. Work with a partner to complete the chart.

Expression	Exponent Law	Equivalent Expression
$8^2 \cdot 8^4$	Product of Powers	8^6
$10^4 \cdot 10^3$	Product of Powers	10^7
$(7^5)^3$	Power of a Power	7^{15}
$(4^8)^4$	Power of a Power	4^{32}
$\frac{2^{12}}{2^{10}}$	Quotient of Powers	2^2
$\frac{6^{20}}{6^{11}}$	Quotient of Powers	6^9
$(3 \cdot 7)^5$	Power of a Product	$3^5 \cdot 7^5$
$(13 \cdot 2)^5$	Power of a Product	$13^5 \cdot 2^5$
$\left(\frac{1}{7}\right)^4$	Power of a Quotient	$\frac{1^4}{7^4}$
$\left(\frac{6}{25}\right)^3$	Power of a Quotient	$\frac{6^3}{25^3}$
$\left(\frac{2}{5}\right)^{-2}$	Negative Exponents	$\frac{5^2}{2^2}$
$11^3 \cdot 11^{-5}$	Product of Powers	$\frac{1}{11^2}$
45^{-1}	Negative Exponents	$\frac{1}{45}$



For question 1, consider having students find the value without a calculator. The square root of question 1 part a, and the cube root of question 1 part b can be done without a calculator. Students may provide both roots for the answer to question 1 part a. Use this to help establish that both roots are correct, but that in the future the principal square root should be considered the answer.



Be strategic on partners and grouping for question 2. Some students will have a strong background knowledge, having seen these exponent laws in a prior course, and some may never have been exposed to these in their course progression. If you have a class full of 8th grade students with limited exposure, consider doing a gradual release into question 2.



Watch for student performance on both sides of the rows. Students may understand how to evaluate the expression using the law, but may not understand how to create the expression from the final value.

Ensure that students understand what an equivalent expression is compared to the value of the expression.

1.4.4 Your Turn! (continued)

3. Marques, Pablo, and Tandice used the product of powers law to write an equivalent expression for $5^{-7} \cdot 5^6$. Each student's equivalent expression is shown in the table.

Marques	Pablo	Tandice
$\frac{1}{5}$	-5	5^{-1}

- a. Which student's answer is not an equivalent expression for $5^{-7} \cdot 5^6$?
Pablo
- b. Which student also applied the negative exponent law?
Marques

4. Find the value of the variable(s) that makes the equation true.

a. $7^{10} = 7^5 \cdot 7^f$ b. $\sqrt[3]{h} \cdot \sqrt[3]{h} \cdot \sqrt[3]{h} = -13$ c. $\sqrt{m} \cdot \sqrt{m} = 26$

$f = 5$ $h = -13$ $m = 26$

d. $\frac{7^5}{6^5} = \left(\frac{6}{7}\right)^y$ e. $35^{12} = (35^x)^2$ f. $\left(\frac{9}{10}\right)^{-1} = \frac{c}{r}$

$y = -5$ $x = 6$ $c = 10$

$r = 9$

g. $18^0 = \frac{18^w}{18^7}$

$w = 7$



- Which exponent law was the easiest for you to recall and use?
- Which exponent law was the most difficult for you to use?



Allow students to share out their answers to question 3 parts a and b. Engage students in a polling protocol where a student pair is asked to share out their answer to the question, and then other students either agree or disagree with the answer and provide justification.



Consider having students articulate and explain how each student arrived at their answer.



For sharing out the answers to question 4, consider infusing physical movement into the activity. For example, post each of the seven expressions at different parts of the room or on different spaces on the board. Group students into seven groups and have each group show their steps on the board (or poster paper) and share with the class how they determined the value, what law they used, and any strategies they used.

1.4.5 Wrap-Up: Cool Down

Component Overview: Students will complete a cool down that has students compare two expressions using inequality symbols. This activity is designed to be done independently to provide data on how each student is doing related to mastery of the content of this lesson.



1.4.5 Wrap-Up: Cool Down

1. Use the symbols $<$, $>$, or $=$ to compare each pair of values. Then, explain your choice.

Expression	Explanation
$2^3 > 2 \cdot 3$	$2^3 = 2 \cdot 2 \cdot 2 = 8$ $2 \cdot 3 = 6$ $8 > 6$
$5^2 < 2^5$	$5^2 = 5 \cdot 5 = 25$ $2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$ $25 < 32$
$100^1 > 1^{100}$	$100^1 = 100$ $1^{100} = 1$ $100 > 1$



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.